IIT ROORKEE

Data-Driven Investment Intelligence Platform

NIFTY-50 Market Data 2000 – 2021

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ABSTRACT

This report presents a comprehensive AI-powered investment intelligence platform built on two decades of NIFTY-50 historical market data. The system integrates four analytical pillars: a machine learning predictor engine (XGBoost and LightGBM), a mean-variance portfolio optimizer for three investor profiles, a multi-metric risk assessment module, and a market anomaly detection framework. The platform achieves 51% directional accuracy consistent with the efficient market hypothesis, constructs portfolios with Sharpe ratios up to 1.20, and identifies 2,600+ anomalous market events aligned with known financial crises. SHAP-based explainability surfaces short-term momentum (Returns_Lag1) and Bollinger Band position as the dominant predictive signals.

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1. Introduction

Financial markets generate enormous volumes of data daily, making it increasingly difficult for investors to extract actionable intelligence from raw price streams. This platform transforms **21 years of NIFTY-50 historical market data** into a multi-layered decision-support system addressing four problems: directional forecasting, portfolio construction, risk quantification, and anomaly detection.

1.1 Dataset Overview

The dataset spans **January 2000 to April 2021** covering **49 NIFTY-50 stocks** across 13 sectors (~380,000 daily OHLCV records). A risk-free rate of **6.0% p.a.** (Indian 10-year G-Sec proxy) and **252 trading days/year** are assumed throughout. INFRATEL is excluded due to truncated history.

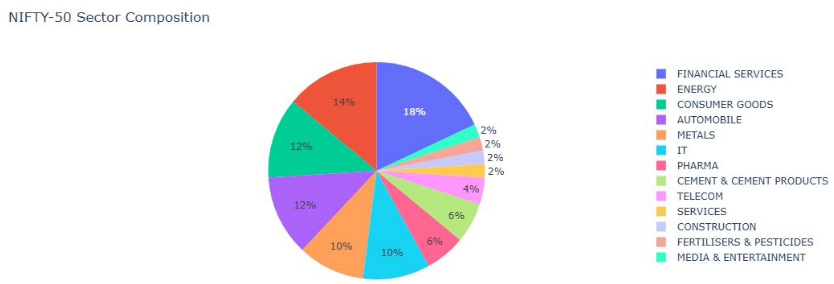

Figure 1. NIFTY-50 Sector Composition. Financial Services (18%) and Energy (14%) are the two largest sectors.

1.2 Preprocessing Pipeline

Each stock's raw CSV is processed via src/preprocess.py: (1) parse and sort DatetimeIndex; (2) forward-fill missing values (max gap: 3 consecutive days); (3) drop residual NaN rows; (4) standardise column names.

2. Exploratory Data Analysis

2.1 Risk–Return Landscape

Figure 2 maps all 49 stocks on the risk–return plane coloured by Sharpe ratio. **SHREECEM** (Sharpe 0.814) and **BAJFINANCE** (0.668) dominate the high-return quadrant. **COALINDIA** is the only stock with a negative annualised return. Defensives (NESTLEIND, POWERGRID) show significantly lower volatility (σ ≈ 0.24–0.30) than the index median (σ ≈ 0.43).

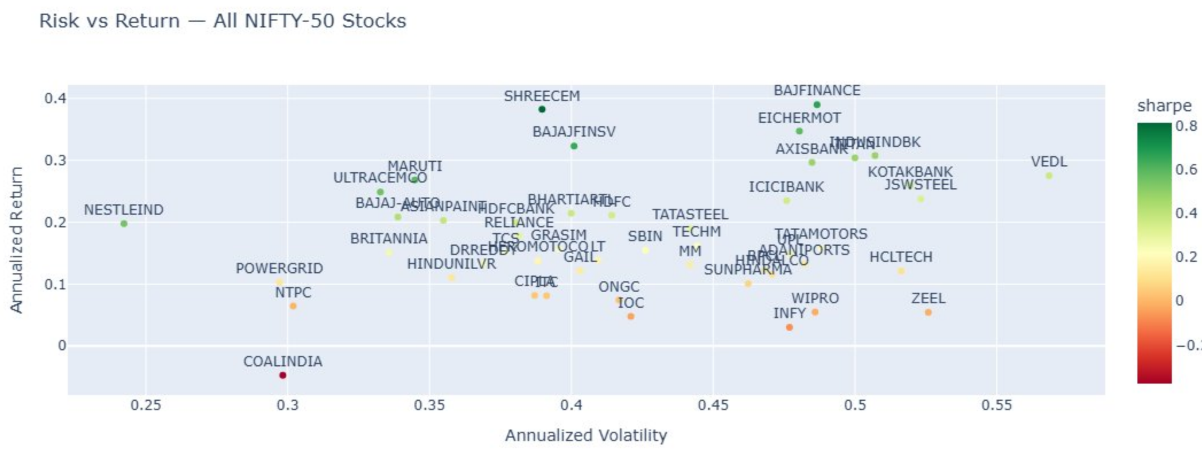

Figure 2. Risk vs Return for all NIFTY-50 constituents (2000–2021). Colour = Sharpe ratio.

2.2 Sector-Level Performance

Cement & Cement Products leads all sectors with average Sharpe ~0.52 (Figure 3). Financial Services follows at ~0.44 despite high volatility, reflecting strong absolute returns. IT and Media & Entertainment rank lowest.

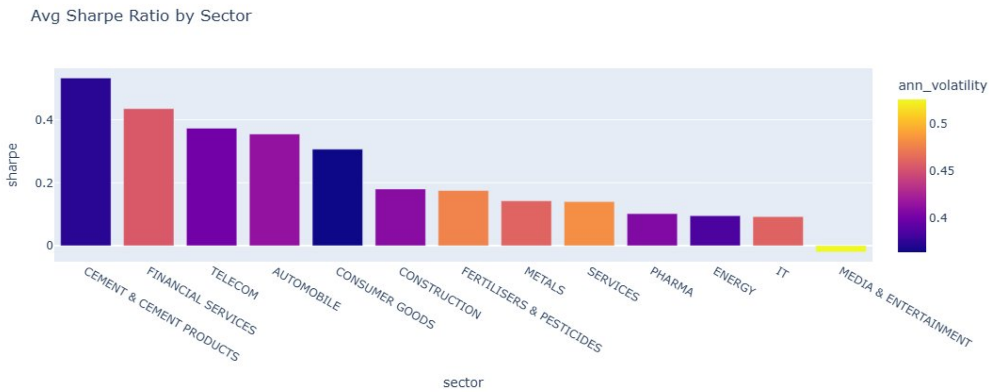

Figure 3. Average Sharpe Ratio by Sector. Darker bar = lower volatility.

2.3 Volume Analysis

Figure 4 shows 20-day MA volume for six representative stocks. Two structural breaks are clearly visible: the 2008 GFC liquidity surge and the March 2020 COVID crash. SBIN and RELIANCE show the most pronounced volume expansion.

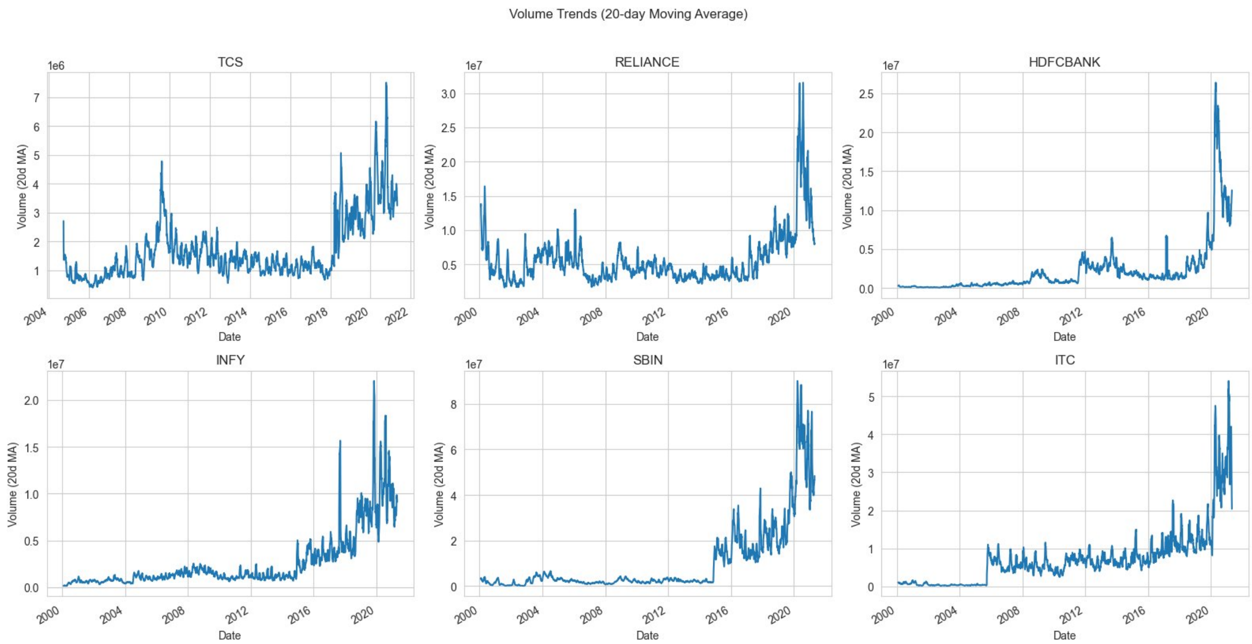

Figure 4. Volume Trends (20-day MA) for TCS, RELIANCE, HDFCBANK, INFY, SBIN, ITC.

3. Feature Engineering

All 18 features are computed in src/indicators.py on single-stock OHLCV DataFrames. No future data is used — all rolling windows are strictly backward-looking.

Category	Features	Description
Trend	SMA 20/50/200, EMA 12/26, Price_vs_SMA	Moving averages; % deviation of close from each SMA
Momentum	RSI, MACD, MACD_Signal, MACD_Hist	14-day RSI; 12/26 EMA crossover with signal line
Volatility	BB_Width, BB_Position, ATR, Volatility_20	Bollinger Bands (20d); ATR (14d); rolling return std dev

Volume	Volume_Ratio	Today's volume / 20-day average volume
Lagged Returns	Returns_Lag1/2/3/5	Daily return lagged 1, 2, 3, 5 days (momentum)
Calendar	DayOfWeek, Month	Day of week (0=Mon); calendar month (1–12)
Target	Target (binary)	1 if next-day close > today's close; else 0

Table 1. Feature set: 18 input features + 1 binary target label.

4. Methodology & Model Architecture

4.1 Problem Formulation

Direction prediction is cast as a **binary classification** problem on a **pooled cross-sectional dataset** of all 49 stocks (~380,000 labelled samples). An **80/20 temporal split** (train on earlier dates, test on the most recent 20%) strictly prevents look-ahead bias.

4.2 Models

XGBoost: 200 estimators, max depth 5, learning rate 0.05, log-loss metric. **LightGBM**: identical hyperparameters, leaf-wise growth. No hyperparameter search performed to avoid overfitting on the temporal hold-out.

4.3 Evaluation Results

Metric	XGBoost	LightGBM
Accuracy	51.0%	51.0%
Precision (Up / Down)	0.51 / 0.51	0.51 / 0.51
Recall (Up / Down)	0.64 / 0.38	0.65 / 0.37
F1 (Up / Down)	0.57 / 0.44	0.57 / 0.43
Directional Accuracy	51.0%	51.0%
Test Samples	45,089	45,089

Table 2. Model performance on held-out temporal test set.

Both models achieve 51% directional accuracy, consistent with the **Efficient Market Hypothesis** for liquid large-cap equities. The higher recall on 'Up' days (0.64–0.65 vs 0.37–0.38) reflects the long-run bullish drift of Indian equities over 2000–2021.

4.4 Feature Importance & SHAP

XGBoost's gain-based importance (Figure 5, left) ranks Month and DayOfWeek highest due to high-purity calendar splits. SHAP (Figure 5, right) gives the true picture: **Returns_Lag1** dominates — short-term momentum is the most learnable signal. BB_Width and BB_Position rank 3rd and 5th, capturing volatility compression before breakouts. This contrast validates SHAP as the primary explainability tool.

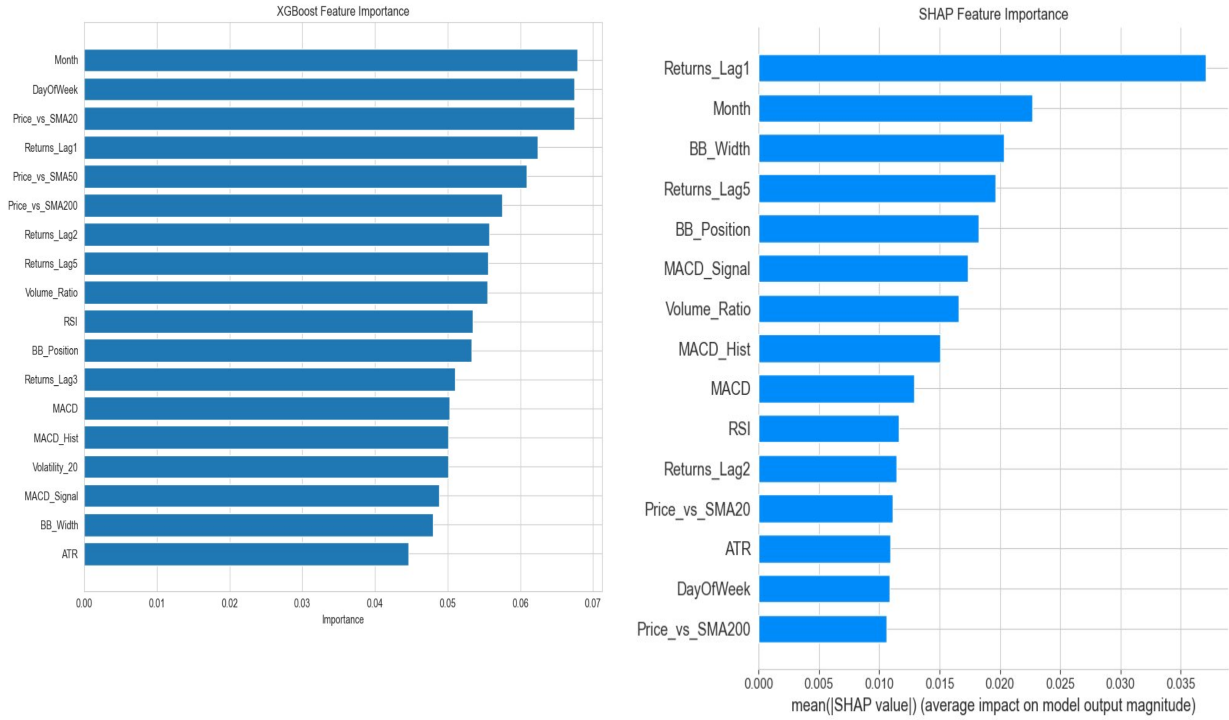

Figure 5. Left: XGBoost gain-based importance. Right: SHAP mean |value|.

5. Portfolio Construction

5.1 Optimisation Methodology

Markowitz Mean-Variance optimisation via SciPy SLSQP using the full 2000–2021 covariance matrix. Per-stock weight caps prevent degenerate single-stock concentration.

- **Conservative**: minimise variance; max 10% per stock → 26 stocks
- **Balanced**: maximise Sharpe ratio; max 20% per stock → 10 stocks
- **Aggressive**: maximise annualised return; max 30% per stock → 4 stocks

Profile	Ann. Return	Ann. Volatility	Sharpe Ratio	No. Stocks
Conservative	12.34%	13.46%	0.43	26
Balanced	29.36%	19.05%	1.20	10
Aggressive	35.21%	26.08%	1.10	4
Equal-Weight	12.19%	17.50%	0.33	49

Table 3. Portfolio metrics (annualised). Risk-free rate = 6.0% p.a. Equal-Weight is the baseline.

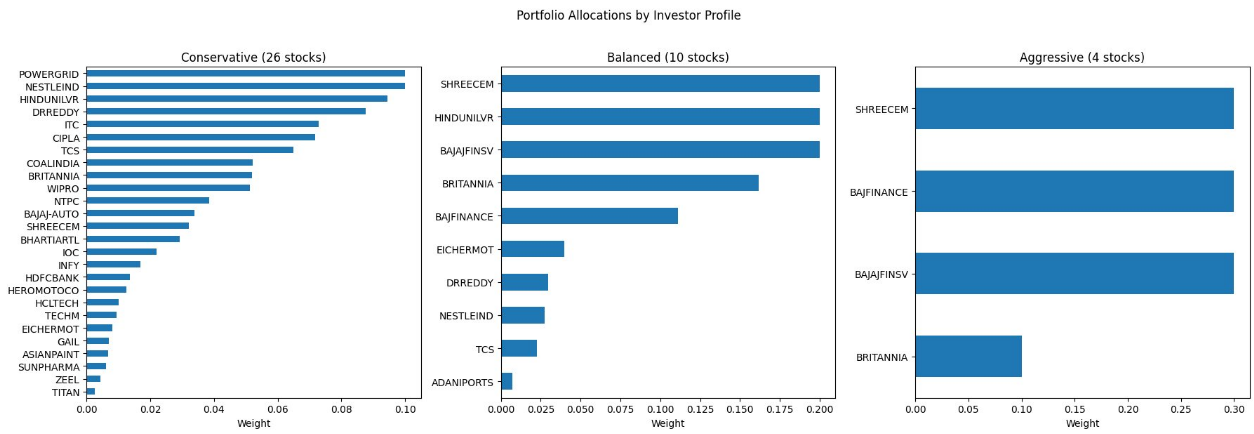

Figure 6. Stock-level allocations: Conservative (26 stocks), Balanced (10), Aggressive (4).

Sector tilts: Conservative → Consumer Goods 32.9%, Energy 16.7% (low-beta defensives). Balanced → Consumer Goods 38.9%, Financial Services 31.1% (best Sharpe/volatility balance). Aggressive → Financial Services 60% (BAJFINANCE + BAJAJFINSV at 30% each), Cement 30% (SHREECEM).

6. Risk Assessment

Five metrics computed per stock across the full history: Annualised Volatility, Sharpe Ratio, Sortino Ratio, Maximum Drawdown, and 95% VaR / CVaR (Expected Shortfall).

Symbol	Return	Vol.	Sharpe	Sortino	Max DD	VaR 95%	Sector
SHREECEM	38.2%	39.0%	0.814	1.277	-80.0%	-3.54%	Cement
BAJFINANCE	39.0%	48.7%	0.668	0.801	-93.3%	-3.85%	Fin. Svcs
BAJAJFINSV	32.3%	40.1%	0.644	0.905	-86.8%	-3.49%	Fin. Svcs
MARUTI	26.8%	34.5%	0.590	0.877	-61.3%	-3.17%	Automobile
EICHERMOT	34.7%	48.0%	0.587	0.729	-93.8%	-3.73%	Automobile
NESTLEIND	19.8%	24.2%	0.549	0.859	-32.4%	-2.15%	Cons. Goods
ULTRACEMCO	24.9%	33.3%	0.553	0.848	-77.8%	-3.06%	Cement
INDUSINDBK	30.8%	50.7%	0.479	0.640	-85.1%	-4.16%	Fin. Svcs
TITAN	30.4%	50.0%	0.478	0.583	-96.7%	-4.05%	Cons. Goods
AXISBANK	29.7%	48.5%	0.478	0.601	-85.0%	-4.16%	Fin. Svcs

Table 4. Risk metrics for top-10 Sharpe stocks. Max DD = Maximum Drawdown.

Key observation: NESTLEIND delivers Sharpe 0.549 with only 24.2% volatility and a maximum drawdown of just 32.4% — the best risk-adjusted defensive profile. Most NIFTY-50 stocks experienced maximum drawdowns exceeding 80% (TITAN: –96.7%, HCLTECH: –96.8%), underscoring the need for drawdown-aware portfolio sizing.

7. Anomaly Detection & Explainability

7.1 Anomaly Detection

Return anomalies are flagged when |Z-score| > 3σ from each stock's historical return distribution. Across all 49 stocks, **2,600+ anomalous events** were detected over 21 years. Top stocks by anomaly count: SHREECEM (80), INDUSINDBK (79), ZEEL (77). Events cluster sharply around the 2008 GFC and March 2020 COVID crash.

Volume spikes use a rolling 60-day Z-score (threshold: 3σ), independently confirming the same crisis windows. **Volatility regimes** classify each day as High (20d/60d vol ratio > 1.5), Normal, or Low (< 0.5) — markets spend the majority of time in the Normal regime.

7.2 SHAP Explainability

SHAP TreeExplainer applied to 500 test samples reveals: **Returns_Lag1** has the highest mean |SHAP| by a wide margin — short-term momentum is the most learnable signal. **Month** ranks second (Indian market seasonality). BB_Width and BB_Position rank 3rd and 5th (volatility compression before breakouts). XGBoost's built-in importance inflates calendar features due to split-purity artefacts; SHAP correctly identifies their lower actual contribution.

7.3 Stock-Level Text Insights

generate_stock_insight() in src/explainability.py generates plain-English summaries: RSI zone, MACD direction, price vs SMA 20/50/200, Bollinger Band position, and 20-day volatility — surfaced in the Streamlit Stock Explorer.

8. Key Findings & Conclusions

- 51% directional accuracy is consistent with EMH for liquid large-caps. Platform value lies in portfolio and risk tools, not standalone forecasting.
- Returns_Lag1 is the dominant SHAP feature — yesterday's return is the most learnable daily signal.
- The Balanced portfolio achieves Sharpe 1.20 at 19.1% volatility, far outperforming the equal-weight baseline (Sharpe 0.33).
- NESTLEIND is the best risk-adjusted compounder: volatility 24.2%, max drawdown –32.4%, Sharpe 0.549.
- Cement & Cement Products leads all sectors by average Sharpe (0.52), driven by SHREECEM (0.814).
- Anomaly detectors independently confirm 2008 GFC and 2020 COVID as peak-stress periods across both return and volume dimensions.
- Max drawdowns >80% are common across high-quality NIFTY-50 names — investors must size positions to survive deep drawdowns.

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End of Report